

Task 4 — Microservices with Message Queue

GitHub: https://github.com/Romko-studik/apz_homework_2/tree/micro_mq

1. Service Registry

Here we see that config server is launching correctly

```
PS D:\UNI_UCU\APZ\apz_homework_2> docker compose up -d
[+] up 11/11
✓ Network apz_homework_2_banking-net Created
✓ Container mongo Healthy
✓ Container config-server Healthy
✓ Container hz1 Created
✓ Container hz2 Created
✓ Container hz3 Created
✓ Container logging-service-1 Healthy
✓ Container logging-service-3 Healthy
✓ Container logging-service-2 Healthy
✓ Container counter-service Healthy
✓ Container facade-service Created
```

Service registry

```
PS D:\UNI_UCU\APZ\apz_homework_2> $registry = Invoke-RestMethod -Uri "http://localhost:8085/services"
>> $registry.registry.'logging-service'
>> $registry.registry.'counter-service'
http://logging-service-1:8001
http://logging-service-3:8001
http://logging-service-2:8001
http://counter-service:8002
```

2. POST 10 transactions

POST requests - we see that it's queued instead of immediately showing balance:

```
PS D:\UNI_UCU\APZ\apz_homework_2> for ($i = 1; $i -le 10; $i++) {
>> Invoke-RestMethod -Method POST -Uri "http://localhost:8080/transaction" -ContentType "application/json" -Body '{"user_id": "user$1", "amount": $($i * 100)}'
>> }

transaction_id      status
-----
ef2c6ca5-d430-41f1-95b6-67da2fda992 queued
626ff277-2b63-4911-8442-fbf9c67bfd4c queued
1d6d320b-d052-443d-ad38-91509219ed52 queued
9c17ecf7-a86d-4324-b342-c7a71e527654 queued
53821345-f2d2-4486-844a-46fbd110da20 queued
0f22894f-d553-4cbd-9a5d-55eeffb8a353 queued
062bdd61-f74e-4dd4-98b8-1bf38a668373 queued
dad7512f-2740-4f7b-815b-a0c243ba892e queued
35b58670-0512-4b23-8c34-c673c8f8a288 queued
5dc28869-af0d-48a7-a240-b78c90bd91ac queued
```

3. Logging service distribution

logging-service-1 console:

```
[Logging-1] Registered with config server as http://logging-service-1:8001
[Logging-1] Stored tx=626ff227-2b63-4911-8442-fbf9c67bfd4c user=user2 amount=+200.00
INFO:      172.20.0.11:49904 - "POST /log HTTP/1.1" 201 Created
[Logging-1] Stored tx=35b58670-0512-4b23-8c34-c673c8f8a288 user=user9 amount=+900.00
INFO:      172.20.0.11:49904 - "POST /log HTTP/1.1" 201 Created
[Logging-1] Stored tx=5dc20869-af0d-48a7-a240-b78c90bd91ac user=user10 amount=+1000.00
INFO:      172.20.0.11:49904 - "POST /log HTTP/1.1" 201 Created
```

logging-service-2 console:

```
[Logging-2] Connected to Hazelcast at hz2:5701
[Logging-2] Registered with config server as http://logging-service-2:8001
[Logging-2] Stored tx=ef2c6ca5-d430-41f1-95b6-67da2fdaf992 user=user1 amount=+100.00
INFO:      172.20.0.11:59074 - "POST /log HTTP/1.1" 201 Created
[Logging-2] Stored tx=1d6d320b-d052-443d-ad38-91509219ed52 user=user3 amount=+300.00
INFO:      172.20.0.11:59074 - "POST /log HTTP/1.1" 201 Created
[Logging-2] Stored tx=0f22904f-d553-4cbd-9a5d-55eefb8a3535 user=user6 amount=+600.00
INFO:      172.20.0.11:59074 - "POST /log HTTP/1.1" 201 Created
[Logging-2] Stored tx=062bdd61-f74e-4dd4-98b8-1bf38a660373 user=user7 amount=+700.00
INFO:      172.20.0.11:59074 - "POST /log HTTP/1.1" 201 Created
```

logging-service-3 console:

```
[Logging-3] Connected to Hazelcast at hz3:5701
[Logging-3] Registered with config server as http://logging-service-3:8001
[Logging-3] Stored tx=9c17ecf7-a86d-4324-b342-c7a71e527654 user=user4 amount=+400.00
INFO:      172.20.0.11:33040 - "POST /log HTTP/1.1" 201 Created
[Logging-3] Stored tx=53821345-f2d2-4486-844a-46fbd110da20 user=user5 amount=+500.00
INFO:      172.20.0.11:33040 - "POST /log HTTP/1.1" 201 Created
[Logging-3] Stored tx=dad7512f-2740-4f7b-815b-a0c243ba892e user=user8 amount=+800.00
INFO:      172.20.0.11:33040 - "POST /log HTTP/1.1" 201 Created
```

4. Counter service queue processing

counter-service console showing queue consumption:

```

INFO:      Started server process [1]
INFO:      Waiting for application startup.
[Counter] Connected to MongoDB at mongodb://mongo:27017
[Counter] Listening on counter-queue
[Counter] Registered with config server as http://counter-service:8002
[Counter] Queue consumer started
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8002 (Press CTRL+C to quit)
[Counter] Processed tx=4e204727-d1be-45c3-8700-4170863dc005 user=user1 amount=+100.00
[Counter] Processed tx=9450c7a6-c6bf-4dfc-a889-2d07744a32d9 user=user2 amount=+200.00
[Counter] Processed tx=f38b8a9b-2a5c-4a2d-a179-cf7235173c97 user=user3 amount=+300.00
[Counter] Processed tx=06f177f3-7013-4474-bdb6-9068261e1cf0 user=user4 amount=+400.00
[Counter] Processed tx=48dfeb19-41dd-4ace-aed5-8f6613eaf678 user=user5 amount=+500.00
[Counter] Processed tx=42d1da28-f918-403f-a70f-82868c818b38 user=user6 amount=+600.00
[Counter] Processed tx=cb63a2d0-ccf8-483d-9496-62e6b6f84d42 user=user7 amount=+700.00
[Counter] Processed tx=a5030e42-563c-4da8-9937-7768fe4ad7f4 user=user8 amount=+800.00
[Counter] Processed tx=56c2ed85-a77a-4f1f-b264-10b624fab1e0 user=user9 amount=+900.00
[Counter] Processed tx=00737a80-100b-4152-a3ec-ee8811f7ccfd user=user10 amount=+1000.00

```

5. GET transactions

GET /accounts — correct balances after queue processed:

```

PS D:\UNI_UCU\APZ\apz_homework_2> Invoke-RestMethod -Uri "http://localhost:8000/accounts"

balances
-----
@{user1=100.0; user2=200.0; user3=300.0; user4=400.0; user5=500.0; user6=600.0; user7=700.0; user8=800.0; user9=900.0; user10=1000.0}

```

GET /user/user1:

```

PS D:\UNI_UCU\APZ\apz_homework_2> Invoke-RestMethod -Uri "http://localhost:8000/user/user1"

user_id balance transactions
-----
user1    100.0  @{transaction_id=4e204727-d1be-45c3-8700-4170863dc005; timestamp=1779336529.4961329; user_id=user1; amount=100.0}

```

6. Fault Tolerance — counter-service paused

POST while counter is paused — transactions still queue successfully:

```

counter-service

transaction_id      status
-----
96485fff-c651-424f-82b1-47f7823b87d7 queued
3564bdaa-1ffb-4ff8-bc27-f95b166e8860 queued

```

GET /accounts while counter is paused — returns null:

```
{
  "balances": null,
  "note": "Counter service unavailable - transactions are queued"
}
counter-service
{
  "balances": {
    "test": 666.0
  }
}
```

After unpausing — counter drains queue, correct balances restored:

```
INFO:      Started server process [1]
INFO:      Waiting for application startup.
[Counter] Connected to MongoDB at mongoddb://mongo:27017
[Counter] Listening on counter-queue
[Counter] Registered with config server as http://counter-service:8002
[Counter] Queue consumer started
INFO:      Application startup complete.
INFO:      Uvicorn running on http://0.0.0.0:8002 (Press CTRL+C to quit)
[Counter] Processed tx=e34b98d2-7053-4adf-9899-ce7bcc21e341 user=test amount=+111.00
[Counter] Processed tx=e63e3d63-7466-4ba1-81a4-ef20c7789a5f user=test amount=+222.00
INFO:      172.20.0.11:53914 - "GET /balances HTTP/1.1" 200 OK
INFO:      172.20.0.11:35948 - "GET /balances HTTP/1.1" 200 OK
INFO:      172.20.0.11:41500 - "GET /balances HTTP/1.1" 200 OK
INFO:      172.20.0.11:46802 - "GET /balances HTTP/1.1" 200 OK
INFO:      172.20.0.11:46802 - "GET /balance/user1 HTTP/1.1" 404 Not Found
[Counter] Processed tx=96485fff-c651-424f-82b1-47f7823b87d7 user=test amount=+111.00
[Counter] Processed tx=3564bdaa-1ffb-4ff8-bc27-f95b166e8860 user=test amount=+222.00
INFO:      172.20.0.11:35744 - "GET /balances HTTP/1.1" 200 OK
```

Analysis

- While counter-service is paused, POST requests still succeed — transactions accumulate in the Hazelcast Queue
- GET requests return null balance since counter-service is unreachable
- After resuming counter-service, it drains all queued transactions and balances become correct
- In conclusion - facade does not need to wait for counter-service, improving resilience and decoupling